

Dealer Hedging Pressure's Forecast on Dark Pool Order Flow

Predicting Institutional Dark Pool Activity through Options Asymmetry Imbalance

Asset Classes

Options, Equity, Off-Exchange Trades

Primary Hypothesis

Options dealer gamma pressure exhibits a statistically significant relationship with dark pool trading activity.

Motivation

Dark-pools are off-exchange trading platforms, aimed to minimize market impact for institutional traders executing large order blocks. Dark pool order flows are reported post-execution, meaning they are invisible to the broad market in real time. The activity opacity creates a disparity in market participation operations and proprietary insights.

Options markets operate through counterparties on the buying and selling side of contracts. Liquidity providers (LPs) sit at the intersection of these market counterparties, required to act for institutional demands. LPs are forced to delta hedge their exposure that results from writing options contracts. The direction and magnitude of this hedging activity can instigate directional pressure on equity pricing and resulting trading volumes.

This project seeks to investigate the validation of LPs net gamma exposure (GEX) asymmetry as a systematic predictor of dark pool activity. If a statistically significant relationship exists, it would suggest that options GEX imbalance serves as a leading indicator of hidden institutional positioning through nuanced, correlated movement.

Objective

This project investigates whether the hedging pressure on liquidity-providers in options markets contains predictive signals into subsequent off-exchange trading activity, specifically within dark pool trading flows. Liquidity-providers are required to continuously delta-hedge their net GEX, generating a measurable asymmetry in order routing. If this identified asymmetry can be systematically linked to dark pool order activity, LP GEX imbalance retains the agency to serve as a leading indicator of hidden liquidity.

Null Hypothesis

Options dealer gamma pressure does not exhibit a statistically significant relationship with subsequent dark pool trading activity.

Alpha Signal:

The alpha signal constructed as a normalized options asymmetry imbalance ratio is defined as:

$$\text{Alpha}(t) = [\text{Net Dealer Gamma Exposure}(t)] / [\text{Total Open Interest}(t)]$$

- **Interpretation:** High Ratio = Unusual Dealer Positioning = Potential Upcoming Institutional Trading Activity

This alpha signal is computed using end-of-day options chain snapshots. The gamma exposure for LPs is computed under the assumption that LPs are short gamma on net option sales and long gamma on net purchases.

Key Assumptions:

- All Dark Pool activity is eventually disclosed to the public through FINRA to allow historically accurate reconstruction and backtesting of the dark-pool volume.
- Observed dark-pool trading reflects institutional demand as opposed to predetermined institutional activity.
- Utilized options market data is effectively consistent and granular across the sample period, supporting reliable GEX calculations.

Methodologies

- OLS regression
 - With lagged predictor variables
 - Quantifies the relationship between LP GEX asymmetry and subsequent dark-pool activity
- Granger Causality Testing
 - Assesses whether GEX imbalance provides a directional predictive power on dark-pool activity
- Robustness Checks
 - Validates signal consistency
 - Alternative gamma computation
 - Sub-period analysis

Technological Components

Data accumulation will be conducted through Python, utilizing yfinance and pandas to scrape and process real-time options data, utilized to calculate GEX. Statistical analysis, alongside backtesting, will be executed in C++ to reduce computational latency while performing large-scale data processing and regression calculations.

Expected Outcome:

A rejected null hypothesis would validate LP GEX asymmetry as a statistically significant predictor of off-exchange trading activity, conducted in dark-pools, to establish LP hedging pressure as a structural alpha signal.